

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Theory of Computation**

Code No: **CSC262** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Mention the transition function of PDA. List the two ways that PDA accepts the string. Convert the following CFG to PDA. $(S \rightarrow AS \mid \epsilon)(B \rightarrow aAb \mid Bb \mid ab)$
2. List any two regular operators. Minimize the following finite state machine using Table Filling algorithm.
3. Define Turing machine as enumerators of strings of a language. Encode the Turing machine $(TM: (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, B\}, \delta, q_0, B, F))$, with input $(w: ba)$ and (δ) is defined as follows. $(\delta(q_0, b) = (q_1, a, R))$, $(\delta(q_1, a) = (q_1, b, R))$, $(\delta(q_2, b) = (q_1, a, R))$, $(\delta(q_1, B) = (q_2, b, L))$

Section B

Attempt any Eight questions[8×5=40]

4. Does machine always refer to hardware? Justify. Define positive closure and kleene closure.
5. What is undecidable problem? Discuss about Post's Correspondence Problem.
6. Define the language of a grammar. For the grammar $(S \rightarrow OS \mid 0 \mid 1 \mid \epsilon)$, show the leftmost derivation for the string 00100 with its parse tree.
7. Define (ϵ) - closure of a state. Differentiate between Moore and Mealy machine.
8. Represent the following regular grammar to finite automata. $(S \rightarrow a \mid aA \mid bB \mid \epsilon)(A \rightarrow aA \mid aS)(B \rightarrow bS \mid \epsilon)$
9. Design the DFA that accepts binary string ending with "00" and show its extended transition function for the string 111000.
10. Convert the following grammar to CNF. $(S \rightarrow AAB), (A \rightarrow aAb \mid \epsilon), (B \rightarrow aB \mid a)$
11. For the following Turing Machine, test whether the string "() (" is accepted or rejected and show it in transition $(XYB(q_0)((q_1, X, R))((q_3, Y, R))(q_1)((q_1, (, R))((q_2, Y, L))((q_0, X, R))((q_1, Y, R))(q_2)((q_2, (, L))((q_0, X, R))((q_2, Y, L))(q_3)((q_3, Y, R))((q_4, B, R))(q_4))$

-
12. Differentiate between Class P and Class NP problem. Mention the transition function of DFA, NFA and ϵ -NFA.